

Enhancing Early Keratoconus Detection With Multimodal Machine Learning: Integrating Tomography, Biomechanics, and Clinical Risk Factors



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- **OBJECTIVE:** To develop and validate a machine learning (ML) diagnostic system that integrates Scheimpflug tomography, corneal biomechanics, and clinical risk factors (CRF) to enhance the early detection of keratoconus (KC).
- **DESIGN:** Prospective, multicenter, cross-sectional study.
- **PARTICIPANTS:** Patients diagnosed with KC and individuals evaluated in preoperative refractive surgery clinics.
- **METHODS:** Demographic, lifestyle, and clinical ophthalmic data, including Pentacam and Corvis ST measurements, were collected from patients with KC and refractive surgery candidates across 5 centers between 2020 and 2024. The dataset was divided into training, validation, internal test, and external test sets. Least absolute shrinkage and selection operator regression was used to identify predictive variables. Six ML models were trained using 4 feature sets: CRF, device-derived parameters, combined features, and selected features.
- **MAIN OUTCOME MEASURES:** Model performance was evaluated using the area under the receiver operating characteristic curve (AUROC).
- **RESULTS:** The dataset included 1035 eyes from 1035 participants across 5 centers: 590 normal controls, 157 eyes with forme fruste keratoconus (FFKC), 143 with subclinical KC, and 145 with clinical KC. For FFKC detection, the post-feature selection CatBoost model

achieved the highest accuracy (AUROC = 0.975), outperforming the combined-feature (AUROC = 0.963), CRF-only (AUROC = 0.856), and device-only models (AUROC = 0.885) in the validation set. This model also outperformed the tomographic and biomechanical index in internal (AUROC = 0.976 vs 0.813; $P = .048$) and external testing (AUROC = 0.952 vs 0.847; $P = .012$). For subclinical and clinical KC, external testing yielded near-perfect performance (AUROC = 0.991 and 1.000, respectively).

- **CONCLUSIONS:** A multimodal ML system integrating CRF, tomography, and biomechanics improved early KC detection, particularly for FFKC. This approach may enhance clinical decision-making and screening for refractive surgery candidates. (Am J Ophthalmol 2025;280:334–346. © 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.)

KERATOCONUS (KC) IS A PROGRESSIVE CORNEAL ectasia characterized by central or paracentral corneal protrusion and stromal thinning, resulting in progressive myopia, irregular astigmatism, and, ultimately, severe visual impairment.¹ Treatments such as intracorneal ring implants^{2,3} and corneal cross-linking can effectively halt disease progression, highlighting the importance of early diagnosis for timely intervention and the preservation of vision.⁴ Early detection is also critical for screening candidates for refractive surgery, as it helps mitigate the risk of postoperative corneal ectasia in individuals predisposed to KC.⁵

KC presents in 3 forms: clinical keratoconus (CKC), subclinical keratoconus (SKC), and forme fruste keratoconus (FFKC). SKC is typically identified by suspicious tomographic findings suggestive of KC, confirmed in the fellow eye. In contrast, FFKC is characterized by normal corneal tomography and slit-lamp examination despite a confirmed KC diagnosis in the contralateral eye.⁶ In recent years, artificial intelligence (AI) has significantly advanced KC diagnostics.^{7,8} However, accurately identifying FFKC remains a challenge for AI models due to the subtle corneal changes

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associated with this condition.^{9,10} This limitation may arise from the predominant reliance on corneal topography, tomography, or biomechanical data in AI model training.¹¹ To enhance diagnostic accuracy, further research is needed to identify distinct disease features and explore the integration of multimodal assessments, incorporating demographic and lifestyle analyses in suspected KC cases.^{10,12}

Previous studies have identified several common risk factors for KC, including eye rubbing,¹³ atopy,¹⁴ body mass index (BMI),¹⁵ socioeconomic status,¹⁶ UV light exposure,¹⁷ sleeping position,¹⁸ and a family history of KC.¹⁹ While differences in the distribution of these clinical risk factors (CRF) between KC and normal populations may aid in disease differentiation, our understanding of how these factors influence machine learning (ML) model performance for KC detection remains limited. Beyond this, little is known about how integrating multi-source data impacts such models. To address these gaps, this study evaluated the diagnostic performance of combining CRF with device-based parameters to identify KC stages. Specifically, we integrated patient risk profiles, Scheimpflug tomography, and biomechanical assessments into an ML system to enhance early KC detection.

METHODS

This prospective multicenter study investigated patients diagnosed with KC and myopia who received treatment at 5 ophthalmic centers in China between 2020 and 2024: Peking University Third Hospital, Liaoning Aier Eye Hospital, Shandong Lunan Eye Hospital, Dalian No. 3 People's Hospital, and the First Affiliated Hospital of Harbin Medical University. The study protocol was approved by the Medical Science Research Ethics Committee of Peking University Third Hospital (Protocol No. M2023859) and adhered strictly to the principles of the Declaration of Helsinki. Written informed consent was obtained from all adult participants or the legal guardians of minors under 18 years of age.

- **PARTICIPANTS:** Participants were categorized into 4 groups: normal, FFKC, SKC, and CKC, defined as follows:

Normal: This group comprised eyes of individuals attending the preoperative refractive surgery clinic, with no slit-lamp findings or tomographic abnormalities and a corrected distance visual acuity of $\geq 20/20$ Snellen equivalent.

FFKC: This group comprised asymptomatic eyes of individuals with clinical KC in 1 eye, exhibiting no slit-lamp abnormalities or tomographic alterations, with a corrected distance visual acuity of $\geq 20/20$ Snellen equivalent, mean keratometry < 47.00 diopters, an inferior-superior (I-S) value ≤ 1.40 ,²⁰ and a keratometry, inferior-superior, and astigmatism index (KISA%) score < 60 .²¹

SKC: This group comprised asymptomatic eyes with suspicious tomographic findings (mild asymmetric bow-tie pattern and/or inferior-superior localized steepening), a normal slit-lamp examination, confirmed KC in the contralateral eye,⁶ and a KISA% score between 60 and 100, with no fixed thresholds for other keratometric values.²¹

CKC: This group comprised eyes presenting with at least 1 positive finding on slit-lamp examination (eg, clinical corneal thinning, Munson's sign, corneal scarring, Vogt's striae, or Fleischer's ring) and/or at least 1 characteristic tomographic sign of KC (eg, decreased thinnest pachymetry, abnormal pachymetry distribution, skewed asymmetric bow-tie pattern, or increased inferior steepness).¹

All evaluations were conducted collaboratively by 2 experienced corneal specialists. For participants in the normal group, 1 eye was randomly selected for inclusion. In patients with KC, only the eye with milder symptoms was included, while the contralateral eye was excluded. Group classification into the FFKC, SKC, or CKC groups was based on the characteristics of the included eye.

The exclusion criteria were as follows: a history of ocular trauma or surgery; corneal ulcers, keratitis, or cataracts; or posterior segment abnormalities such as uveitis, retinitis, retinal detachment, retinal vascular occlusion, retinal dystrophies, maculopathy, or optic nerve disorders. Participants with severe systemic diseases were also excluded. These excluded diseases comprised autoimmune disorders (eg, systemic lupus erythematosus, Sjögren's syndrome), metabolic conditions (eg, diabetes mellitus, thyroid dysfunction), chronic infectious diseases (eg, HIV, hepatitis), connective tissue disorders (eg, Marfan syndrome, Ehlers-Danlos syndrome), advanced cardiovascular disease, and active malignancy.

- **QUESTIONNAIRE SURVEY:** A structured, face-to-face survey was conducted by trained research staff during participants' initial hospital visit. We developed a standardized questionnaire protocol based on prior KC epidemiological studies^{13,22} and piloted it at 2 centers before full implementation. All staff involved in questionnaire administration underwent standardized training, and routine quality checks were conducted to minimize inter-center variability. The unified demographic questionnaire collected information on age, sex, education level, occupational status, childhood residence, childhood family socioeconomic status, history of eye rubbing, sleeping position, snoring and sleep apnea, UV exposure, screen time and reading habits, life stress, smoking and alcohol consumption, allergic disease history, family history of KC, and personal history of ocular and systemic diseases.

- **INSTRUMENTAL EXAMINATION:** All participants completed a standardized, comprehensive ophthalmologic examination during their first clinic visit, including visual

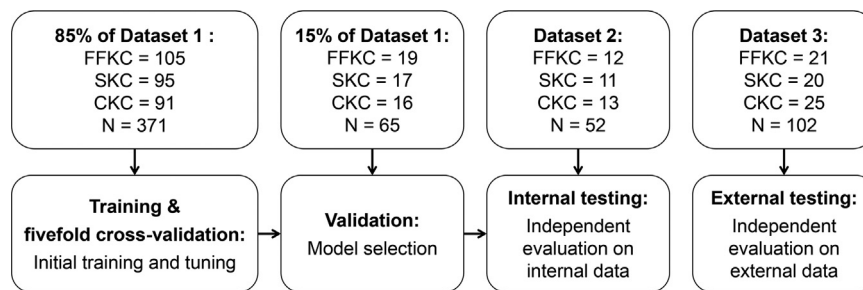


FIGURE 1. Workflow diagram illustrating the training, validation, and testing process based on 3 datasets. CKC = clinical keratoconus; FFKC = forme fruste keratoconus; N = normal corneas; SKC = subclinical keratoconus.

acuity assessment, slit-lamp examination, funduscopy, intraocular pressure measurement, autorefractometry (Topcon RM 8800, Topcon Corporation, Tokyo, Japan), phoropter testing (Topcon CV-5000, Topcon Corporation, Tokyo, Japan), and trial lens evaluation. Additionally, anterior segment optical coherence tomography (Visante, Zeiss, Oberkochen, Germany), corneal tomography (Pentacam, Oculus, Washington, DC, USA), and biomechanical assessments using the Corvis ST device (Oculus; Optikgeräte GmbH, Wetzlar, Germany) were performed. All examinations were performed by trained and certified technicians using identical diagnostic equipment models across all study centers. Regular equipment calibration and procedural audits were conducted to ensure consistency and good data quality. Participants were required to discontinue wearing soft contact lenses for at least 2 weeks and rigid gas-permeable lenses for at least 4 weeks before undergoing testing.

Height and weight were measured with participants wearing light clothing and no shoes using a wall-mounted stadiometer and an electronic scale, respectively. BMI was then calculated using the standard formula: weight (kg)/height (m²).

• **DATA PREPROCESSING:** The built-in normative databases of the Pentacam and Corvis ST devices were used to generate standard indices. Two categories of parameters exported from the Pentacam and Corvis ST devices were excluded from the analysis. The first category included parameters used for study group classification, such as the I-S value and KISA%. The second category comprised existing machine-derived parameters, including the Pentacam topographical KC classification, the Belin-Ambrósio deviation index (BADD), the Corvis biomechanical index (CBI), and the tomographic and biomechanical index (TBI).

Three datasets were used for analysis. Dataset 1, representing the original data, and Dataset 2, used for internal testing, were obtained from Peking University Third Hospital, whereas Dataset 3, designated for external testing, included data from 4 additional centers. Preprocessing steps involved standardizing continuous variables, ap-

plying ordinal encoding for ordered categorical variables, and implementing one-hot encoding for nominal categorical variables. Missing values were imputed using the mean.

• **MACHINE LEARNING:** Dataset 1 was divided into a training set (85%) and a validation set (15%). Six ML models were developed using the training set: logistic regression, random forest (RF), support vector machine (SVM), extreme gradient boosting (XGBoost), categorical boosting (CatBoost), and light gradient boosting machine (LightGBM). To address class imbalance, the synthetic minority oversampling technique (SMOTE) was applied.²³ Hyperparameters were optimized using Bayesian optimization, and fivefold cross-validation was performed on the training set to identify optimal configurations.

Each model was trained using 3 different feature sets: demographic and risk factor features, device-derived features (Pentacam and Corvis ST), and all features combined. The least absolute shrinkage and selection operator was applied to filter features within the training set.²⁴ The final models, based on the selected features, were evaluated on the validation set, internal test set, and external test set. The ML workflow based on these 3 datasets is illustrated in Figure 1.

The models generated both deterministic and probabilistic outputs. Performance was assessed for detecting FFKC, SKC, CKC, and all KC cases combined vs normal eyes. Model performance was evaluated using accuracy, sensitivity, specificity, F1 score, area under the receiver operating characteristic curve (AUROC), area under the precision-recall curve, and Brier score. Accuracy, sensitivity, and specificity were reported with 95% CIs, calculated using the Wilson method.²⁵ Pairwise AUROC comparisons were conducted using the DeLong test,²⁶ with statistical significance set at $P < .05$.

• **STATISTICAL ANALYSIS:** All statistical analyses were performed using the SPSS 25.0 software package (SPSS Inc, Chicago, IL, USA). Categorical variables were presented as event counts, while continuous variables were expressed as means $\pm$ SDs. The normality of continuous variables was assessed using the Shapiro–Wilk test. Since at least 1 group

for each variable did not follow a normal distribution, the nonparametric Kruskal–Wallis test was used for intergroup comparisons. The Chi-squared test was used for categorical data. Post-hoc analyses, including Dunn’s test or Bonferroni correction, were used to adjust for multiple comparisons.

RESULTS

• **BASELINE CHARACTERISTICS:** This study analyzed 1035 eyes from 1035 participants across 3 datasets, comprising 779 from Dataset 1, 88 from Dataset 2, and 168 from Dataset 3. Among these, 590 were classified as normal, 157 as FFKC, 143 as SKC, and 145 as CKC. No significant differences were observed in age ($P = .603$) or right/left eye distribution ($P = .083$) among the 4 groups. Representative Pentacam and Corvis ST measurements for each category are presented in Figure 2. Detailed baseline characteristics are presented in Table 1. Additionally, no significant differences in baseline characteristics were found among the 3 datasets within each diagnostic category (all $P > .05$).

Key demographic, lifestyle, and selected examination parameters were analyzed collectively across all 3 datasets. Significant differences were identified between the control and KC groups in BMI, sex, eye-rubbing intensity, frequency, and duration, as well as education level, childhood residence, and early-life socioeconomic status. However, no significant differences were observed among the KC subgroups. The prevalence of allergic conjunctivitis was higher in patients with KC than in controls, although this difference did not reach statistical significance (Figure 3).

All examination parameters differed significantly among groups ($P < .001$ for all comparisons). Morphological and biomechanical disparities were particularly pronounced between CKC and control eyes. Maximum keratometry and stress-strain index did not significantly differ between FFKC and normal eyes (both $P > .99$), whereas other parameters exhibited marked distinctions (Figure 4).

• **FEATURE SELECTION:** From the combined dataset, 111 features were initially identified, comprising 31 CRF parameters, 52 Pentacam parameters, and 28 Corvis ST parameters. Using data from 662 patients in the training set, this feature set was refined to 75 potential predictors, comprising 25 CRF, 30 Pentacam, and 20 Corvis ST parameters (Figure 5A). The initial feature list and the selection status of individual CRF, Pentacam, and Corvis ST parameters are presented in Supplementary Tables 1–3.

• **MODEL PERFORMANCE ACROSS FEATURE COMBINATIONS:** The 6 ML algorithms were evaluated for their ability to differentiate FFKC from normal cases using various feature combinations on the validation set (Table 2). The CatBoost and XGBoost models demonstrated the highest performance using either CRF-only or examination-only

parameters, achieving AUROCs of 0.856 and 0.892, respectively. Incorporating all features improved diagnostic accuracy, with CatBoost attaining an AUROC of 0.963, significantly outperforming the best-performing CRF-only ($P = .010$, DeLong test) and examination-only ($P = .035$) models. Regarding calibration, the full-feature CatBoost model demonstrated superior performance, yielding a lower Brier score (0.076) compared with the CRF-only (0.105) and examination-only (0.183) models. After feature selection, model performance remained robust, with CatBoost achieving a peak AUROC of 0.975 (Figure 5B) and an improved Brier score of 0.055.

For SKC and CKC classification, CatBoost models trained on CRF features achieved AUROCs of 0.820 and 0.849, respectively, which were comparable to the FFKC classification results. However, examination parameters alone yielded near-perfect accuracy for SKC and CKC detection (AUROCs of 0.999 and 1.000, respectively). No significant differences in performance were observed between full-featured, selected, or examination-only models for SKC and CKC classification (Figure 5C and D), with Brier scores ranging from 0.012 to 0.028.

• **COMPARISON OF EXISTING PARAMETERS AND THE FINAL MODEL:** Receiver operating characteristic analysis was performed to compare the performance of BADD, CBI, TBI, and the final CatBoost model based on selected features across internal and external test sets (Table 3). While existing parameters effectively differentiated SKC and CKC from normal corneas, their performance declined significantly in detecting FFKC. In contrast, the final model demonstrated consistent accuracy across all classifications (Figure 6) and showed stable calibration, with Brier scores ranging from 0.015 to 0.080.

For FFKC detection, the model significantly outperformed TBI in both internal testing (AUROC = 0.976 vs 0.813; $P = .048$, DeLong test) and external testing (AUROC = 0.952 vs 0.847; $P = .012$). However, no significant differences were observed between the model and existing parameters for SKC and CKC detection in either test set.

DISCUSSION

The findings of this study highlight the potential of a multimodal ML approach that integrates Scheimpflug tomography, corneal biomechanics, and CRF for the early detection of KC. Our results demonstrated that combining these data sources enhances diagnostic accuracy, particularly in detecting FFKC, which remains challenging for conventional diagnostic methods.

Currently, there is no standardized criterion for defining early KC.⁶ Some studies classify early ectasia into asymmetric eyes with normal topography (or SKC) and very asymmetric eyes with normal topography and tomography (or

TABLE 1. Baseline Participant Characteristics

Parameter	Dataset 1				Dataset 2				Dataset 3			
	N	FFKC	SKC	CKC	N	FFKC	SKC	CKC	N	FFKC	SKC	CKC
No. of eyes	436	124	112	107	52	12	11	13	102	21	20	25
Eye, L	219 (50.2)	66 (53.2)	59 (52.7)	47 (43.9)	25 (48.1)	6 (50.0)	6 (54.5)	7 (53.8)	51 (50.0)	11 (52.4)	9 (45.0)	12 (48.0)
Sex, M	235 (53.9)	98 (79.0)	76 (67.9)	78 (72.9)	28 (53.8)	8 (66.7)	7 (63.6)	7 (53.8)	41 (40.2)	13 (61.9)	11 (55.0)	14 (56.0)
Age, years	26.6 (7.1)	25.0 (5.8)	26.6 (7.3)	27.7 (7.5)	27.0 (7.0)	22.3 (5.2)	26.7 (7.6)	24.5 (6.4)	26.0 (6.6)	25.6 (6.7)	27.7 (7.5)	27.3 (7.0)
BMI, kg/m ²	22.35 (3.13)	23.89 (3.94)	23.87 (3.95)	22.95 (3.48)	22.49 (5.34)	23.55 (5.17)	23.43 (5.21)	22.58 (6.38)	22.36 (5.13)	23.96 (5.77)	24.02 (6.07)	23.65 (5.56)
Kmax, D	44.58 (1.73)	44.59 (1.60)	47.53 (2.47)	55.42 (7.71)	44.54 (1.44)	45.48 (1.99)	46.22 (1.41)	53.50 (4.89)	44.57 (1.64)	44.71 (1.55)	47.38 (3.87)	60.63 (10.14)
TCT, μ m	543.61 (28.57)	516.59 (33.50)	489.46 (37.27)	446.27 (39.99)	540.90 (22.23)	515.83 (28.79)	489.36 (37.14)	445.23 (23.45)	542.69 (26.27)	505.86 (25.28)	489.95 (40.58)	434.32 (36.76)
I-S value, D	0.28 (0.63)	0.37 (0.69)	1.95 (0.87)	6.13 (3.24)	0.02 (0.64)	0.59 (0.74)	1.71 (0.71)	5.13 (2.04)	0.16 (0.62)	0.49 (0.61)	1.96 (0.87)	5.78 (2.99)
KISA%	8.08 (26.03)	12.66 (14.60)	38.51 (32.35)	1706.36 (3095.62)	6.05 (6.00)	6.59 (5.94)	40.60 (44.08)	1052.74 (1894.42)	7.03 (9.54)	11.61 (11.99)	39.52 (42.39)	1822.38 (3623.78)
biOP, mmHg	16.35 (1.94)	15.17 (1.86)	14.93 (2.16)	14.60 (2.33)	16.24 (2.93)	16.18 (2.08)	14.55 (1.50)	14.34 (2.20)	16.22 (2.23)	15.50 (1.52)	14.90 (1.82)	14.02 (2.34)
DAR2	3.97 (0.48)	4.30 (0.44)	4.67 (0.65)	5.62 (1.44)	4.27 (1.97)	4.23 (0.51)	4.79 (0.55)	5.32 (0.50)	4.05 (0.62)	4.36 (0.41)	4.56 (0.64)	6.51 (1.49)
SPA1, Hg/mm	119.30 (16.58)	102.70 (17.29)	89.44 (19.77)	68.46 (18.72)	126.93 (16.17)	102.77 (14.90)	91.65 (13.57)	69.24 (11.53)	123.57 (15.02)	104.23 (13.51)	87.60 (22.23)	60.12 (19.72)
SSI	0.81 (0.13)	0.80 (0.13)	0.77 (0.15)	0.68 (0.15)	0.82 (0.15)	0.81 (0.16)	0.76 (0.12)	0.70 (0.05)	0.80 (0.12)	0.85 (0.10)	0.87 (0.12)	0.57 (0.06)

Continuous data are presented as means (standard deviation) and categorical variables as numbers (percentages).
N = normal corneas, FFKC = forme fruste keratoconus, SKC = subclinical keratoconus, CKC = clinical keratoconus, L = left, M = male, BMI = body mass index, Kmax = maximum keratometry, D = diopter, TCT = thinnest corneal thickness, I-S = inferior-superior, KISA% = keratometry, inferior-superior, and astigmatism index, biOP = biomechanically corrected intraocular pressure, DAR2 = deformation amplitude measured at 2 mm from the center, SPA1 = stiffness parameter at first applanation, and SSI = stress-strain index.

TABLE 2. Statistical Comparison of Machine Learning Models Using Different Feature Combinations for Distinguishing Forme Fruste Keratoconus From Normal Cases in the Validation Set

Model	Accuracy (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	F ₁ Score	AUROC	AUPRC	Brier Score
CRF							
LR	0.738 (0.635-0.820)	0.737 (0.512-0.882)	0.739 (0.621-0.830)	0.560	0.821	0.585	0.179
RF	0.750 (0.648-0.830)	0.737 (0.512-0.882)	0.754 (0.637-0.842)	0.571	0.832	0.511	0.172
SVM	0.714 (0.610-0.800)	0.737 (0.512-0.882)	0.708 (0.588-0.804)	0.539	0.829	0.605	0.186
XGBoost	0.786 (0.687-0.860)	0.790 (0.567-0.915)	0.785 (0.670-0.867)	0.625	0.835	0.598	0.166
CatBoost	0.881 (0.795-0.934)	0.632 (0.410-0.809)	0.954 (0.873-0.984)	0.706	0.856	0.742	0.105
LightGBM	0.810 (0.713-0.879)	0.790 (0.567-0.915)	0.815 (0.705-0.891)	0.652	0.850	0.717	0.149
Pentacam + Corvis ST							
LR	0.905 (0.823-0.951)	0.684 (0.460-0.846)	0.969 (0.895-0.992)	0.765	0.859	0.741	0.094
RF	0.821 (0.726-0.889)	0.895 (0.686-0.971)	0.800 (0.687-0.879)	0.694	0.883	0.698	0.167
SVM	0.774 (0.674-0.850)	0.947 (0.754-0.991)	0.723 (0.604-0.817)	0.655	0.885	0.617	0.176
XGBoost	0.798 (0.700-0.870)	0.790 (0.567-0.915)	0.800 (0.687-0.879)	0.638	0.892	0.603	0.154
CatBoost	0.714 (0.610-0.800)	0.842 (0.624-0.945)	0.677 (0.556-0.778)	0.571	0.873	0.763	0.183
LightGBM	0.881 (0.795-0.934)	0.737 (0.512-0.882)	0.923 (0.832-0.967)	0.737	0.866	0.791	0.109
CRF + Pentacam + Corvis ST							
LR	0.917 (0.838-0.959)	0.895 (0.686-0.971)	0.923 (0.832-0.967)	0.829	0.949	0.906	0.078
RF	0.905 (0.823-0.951)	0.632 (0.410-0.809)	0.985 (0.918-0.997)	0.750	0.926	0.850	0.083
SVM	0.881 (0.795-0.934)	0.790 (0.567-0.915)	0.908 (0.813-0.957)	0.750	0.952	0.888	0.074
XGBoost	0.917 (0.838-0.959)	0.790 (0.567-0.915)	0.954 (0.873-0.984)	0.811	0.952	0.886	0.067
CatBoost	0.905 (0.823-0.951)	0.684 (0.460-0.846)	0.969 (0.895-0.992)	0.765	0.963	0.887	0.076
LightGBM	0.905 (0.823-0.951)	0.790 (0.567-0.915)	0.939 (0.852-0.976)	0.790	0.940	0.860	0.088
CRF + Pentacam + Corvis ST (after feature selection)							
LR	0.893 (0.809-0.943)	0.790 (0.567-0.915)	0.923 (0.832-0.967)	0.769	0.967	0.913	0.066
RF	0.881 (0.795-0.934)	0.579 (0.363-0.769)	0.969 (0.895-0.992)	0.688	0.893	0.777	0.097
SVM	0.869 (0.781-0.925)	0.842 (0.624-0.945)	0.877 (0.776-0.936)	0.744	0.944	0.889	0.079
XGBoost	0.917 (0.838-0.959)	0.790 (0.567-0.915)	0.954 (0.873-0.984)	0.811	0.970	0.914	0.065
CatBoost	0.929 (0.853-0.967)	0.790 (0.567-0.915)	0.969 (0.895-0.992)	0.833	0.975	0.921	0.055
LightGBM	0.929 (0.853-0.967)	0.737 (0.512-0.882)	0.985 (0.918-0.997)	0.824	0.962	0.902	0.067

AUPRC = area under the precision-recall curve, AUROC = area under the receiver operating characteristic curve, CatBoost = categorical boosting, CI = confidence interval, CRF = clinical risk factors, LightGBM = light gradient boosting machine, LR = logistic regression, RF = random forest, SVM = support vector machine, XGBoost = extreme gradient boosting.

TABLE 3. Performance of the Final Model Based on the Selected Features

Model	Accuracy (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	F ₁ Score	AUROC	AUPRC	Brier Score
Internal test set							
FFKC vs N	0.922 (0.830-0.966)	0.917 (0.646-0.985)	0.923 (0.818-0.970)	0.815	0.976	0.899	0.065
SKC vs N	1.000 (0.943-1.000)	1.000 (0.741-1.000)	1.000 (0.931-1.000)	1.000	1.000	1.000	0.015
CKC vs N	0.969 (0.895-0.992)	1.000 (0.772-1.000)	0.962 (0.870-0.989)	0.929	1.000	1.000	0.038
KC vs N	0.932 (0.859-0.968)	0.917 (0.782-0.971)	0.942 (0.844-0.980)	0.917	0.986	0.983	0.042
External test set							
FFKC vs N	0.919 (0.857-0.955)	0.762 (0.549-0.894)	0.951 (0.890-0.979)	0.762	0.952	0.858	0.061
SKC vs N	0.934 (0.876-0.966)	0.950 (0.764-0.991)	0.931 (0.865-0.966)	0.826	0.991	0.975	0.061
CKC vs N	0.921 (0.861-0.957)	1.000 (0.867-1.000)	0.902 (0.829-0.946)	0.833	1.000	1.000	0.059
KC vs N	0.881 (0.823-0.922)	0.864 (0.761-0.927)	0.892 (0.817-0.939)	0.851	0.963	0.959	0.080

AUPRC = area under the precision-recall curve, AUROC = area under the receiver operating characteristic curve, CI = confidence interval, N = normal corneas, FFKC = forme fruste keratoconus, SKC = subclinical keratoconus, CKC = clinical keratoconus, KC = keratoconus.

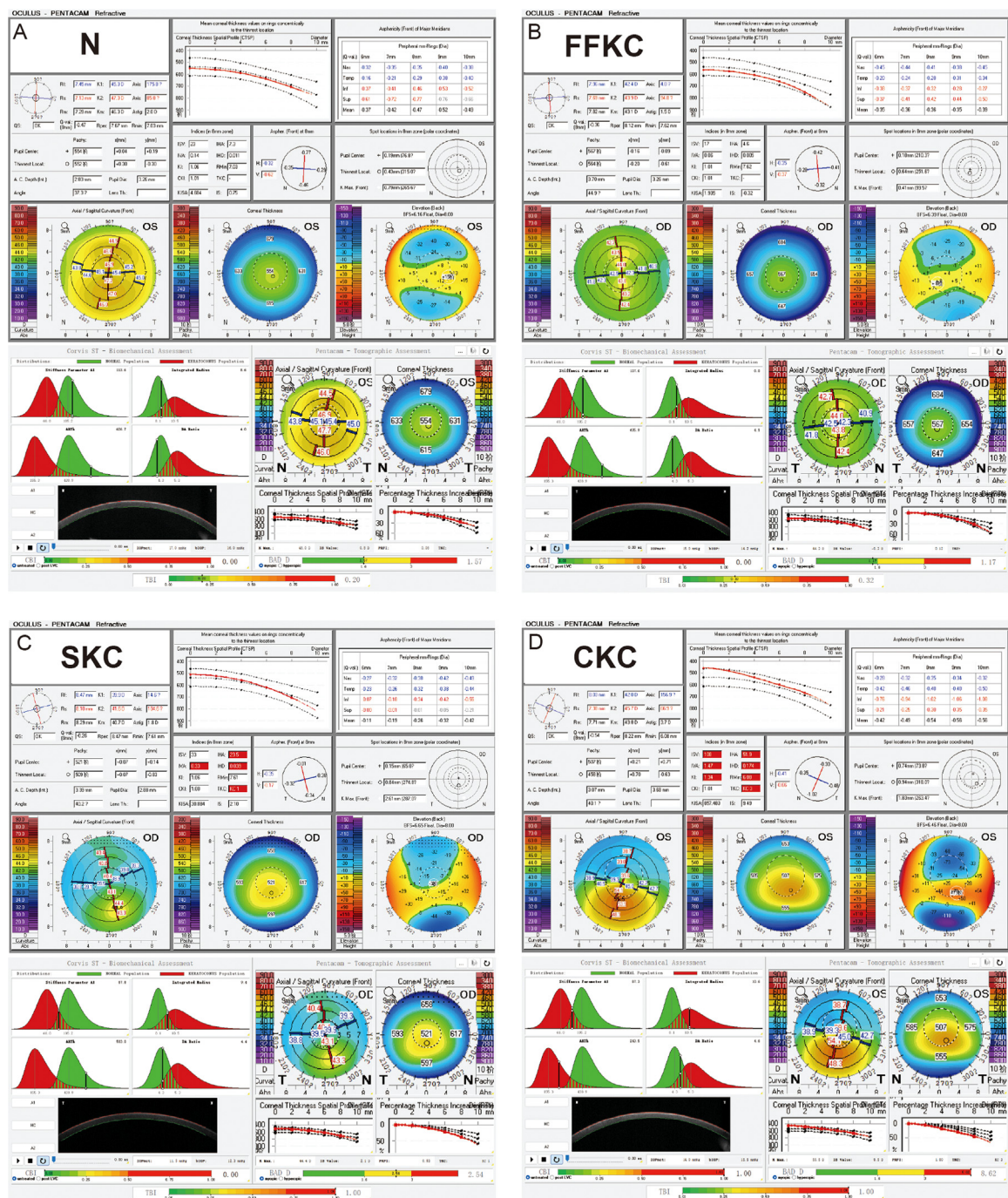


FIGURE 2. Example measurements obtained with pentacam (Upper images) and corvis st (Lower Images). A. Normal corneas. B. Forme fruste keratoconus. C. Subclinical keratoconus. D. Clinical keratoconus.

FFKC).²⁷ In this study, we adopted the terms “FFKC” and “SKC” for early KC with explicit classification criteria.²⁸ Despite variability in definitions, the FFKC group provides a critical framework for evaluating parameters that quantify predisposition to corneal biomechanical decompensation.²⁸⁻³⁰

FFKC eyes exhibited significant alterations in certain examination parameters, such as stiffness at first applan-

tion, consistent with previous research.³¹ These metrics show promise for distinguishing FFKC from normal eyes. However, significant overlap in parameter values between groups complicates definitive diagnosis.²⁸ The advent of multimodal AI has marked a significant advancement in FFKC detection. Recent studies have integrated data from Scheimpflug and Placido imaging³² or combined tomography and biomechanics³³ to improve early KC diagno-

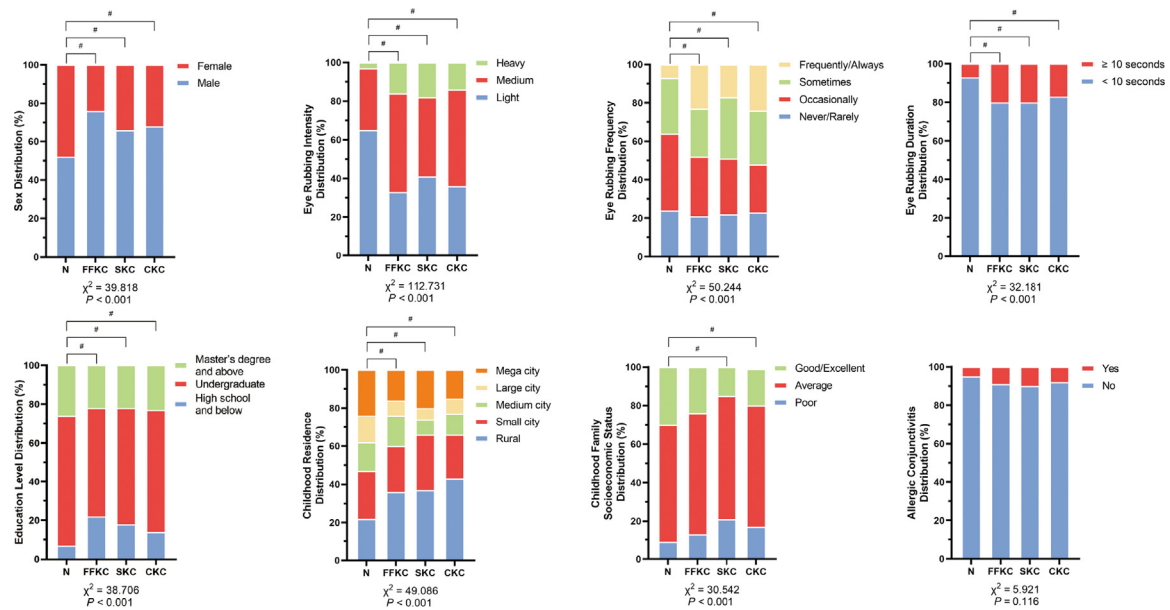


FIGURE 3. Stacked bar charts illustrating the demographic characteristics and lifestyle distribution of patients across all groups. # Statistically significant difference based on Bonferroni correction. CKC = clinical keratoconus; FFKC = forme fruste keratoconus; N = normal corneas; SKC = subclinical keratoconus.

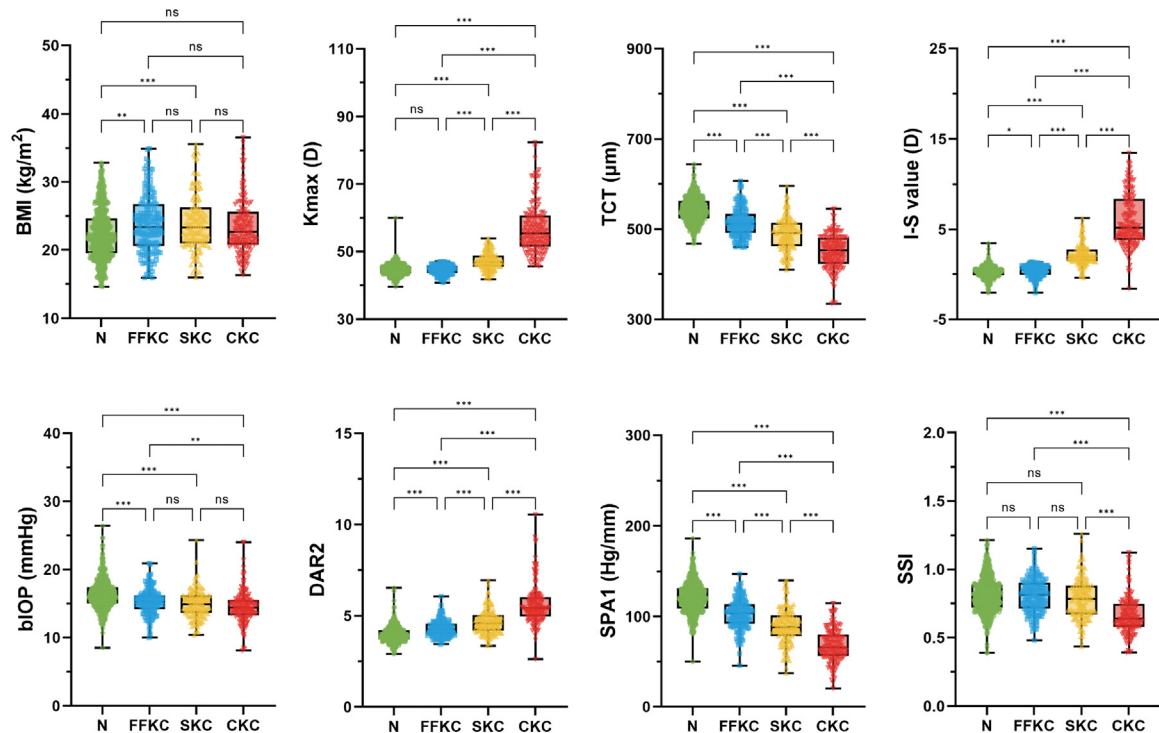


FIGURE 4. Boxplots illustrating BMI, pentacam, and corvis ST parameters across all groups. *P < .05, **P < .01, and ***P < .001. bIOP = biomechanically corrected intraocular pressure; BMI = body mass index; CKC = clinical keratoconus; D = diopter; DAR2 = deformation amplitude measured at 2 mm from the center; FFKC = forme fruste keratoconus; I-S = inferior-superior index; Kmax = maximum keratometry; N = normal corneas; SKC = subclinical keratoconus; SPA1 = stiffness parameter at first applanation; SSI = stress-strain index; TCT = thinnest corneal thickness.

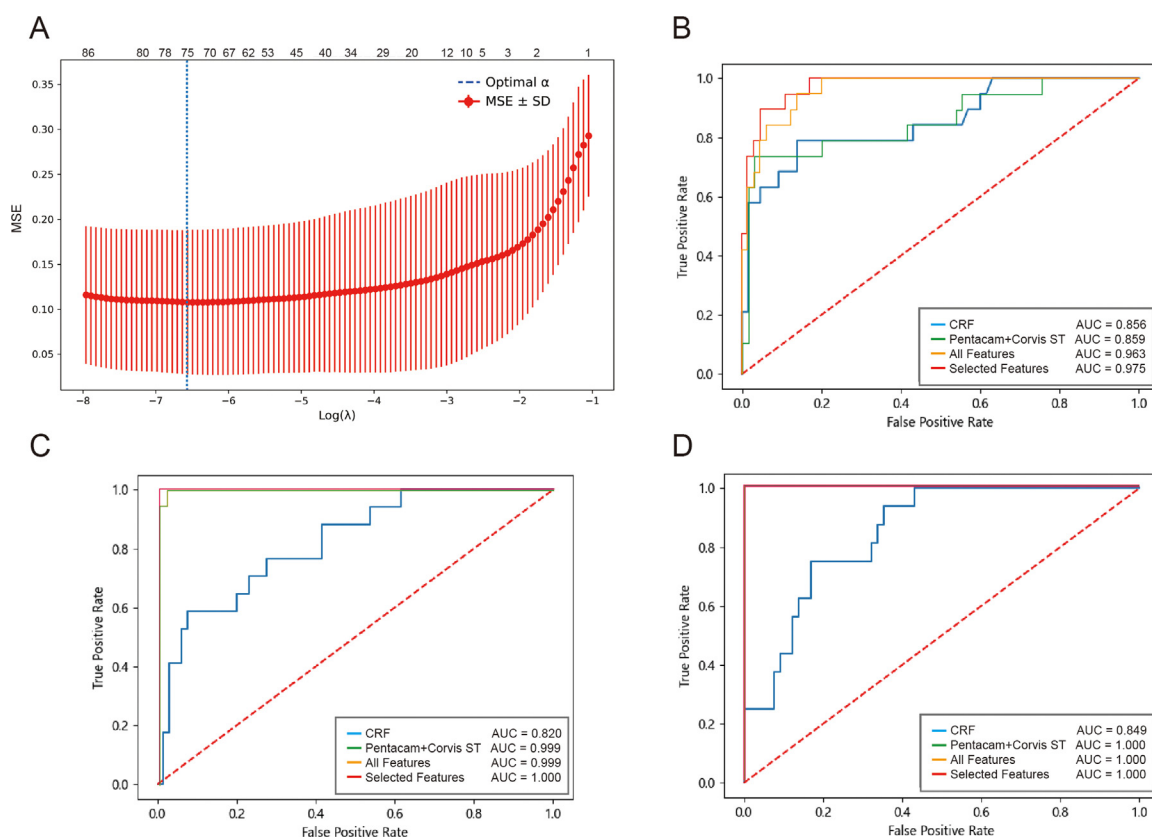


FIGURE 5. Feature selection and comparative model performance evaluation. **A.** Least absolute shrinkage and selection operator logistic regression plot. **B-D.** Receiver operating characteristic analysis of categorical boosting models with different feature combinations for distinguishing forme fruste, subclinical, and clinical keratoconus from normal cases in the validation set. AUC = area under the curve; CRF = clinical risk factors; MSE = mean squared error; SD = SD.

sis. While ML-based parameters derived from single-source data, such as BADD and CBI, have demonstrated high accuracy in diagnosing SKC and CKC, their performance is less reliable for FFKC. In contrast, indices incorporating data from multiple devices, such as TBI, show greater differentiation of FFKC, though further refinement is necessary.³⁴

Given the subtle corneal changes associated with FFKC, diagnostic performance based solely on machine-measured parameters and models remains limited. The need to go beyond corneal shape and biomechanical assessment to define ectasia risk has been sustained and supported.¹⁰ Established KC risk factors, including eye rubbing,¹³ atopy,¹⁴ BMI,¹⁵ socioeconomic status,¹⁶ and a family history of KC,¹⁹ were corroborated in this study. Compared with the control group, patients with KC, including those with FFKC, exhibited a higher prevalence of male sex, elevated BMI, lower educational attainment, rural childhood residence, poorer childhood economic status, and intensified eye-rubbing habits. These factors, independent of disease severity, suggest that CRF integration could refine ML models by capturing holistic health profiles and ectasia susceptibility. However, it is important to acknowledge that this analysis was conducted exclusively within a Chinese pop-

ulation. Although some risk factors, such as eye rubbing, are consistently recognized across ethnicities, with similar prevalence estimates ranging from 44% to 66% in both Asian and white populations,³⁵ the overall prevalence of KC and the influence of specific risk factors vary by geographic region.³⁶ For example, atopy has been reported in only 6.5% of Indian KC patients, while rates can be as high as 22% in white and Arab populations. Similarly, the proportion of patients with a family history of KC ranges from 5% in Scotland to 28% in Turkey, reflecting the genetic and environmental heterogeneity of the disease.³⁵ These disparities suggest the possible existence of KC subtypes with distinct etiological pathways across populations. While our findings provide valuable insight into CRF within the Chinese demographic, further studies involving ethnically and geographically diverse populations are essential to validate and enhance the generalizability of the proposed model.

Previous studies have developed AI models for disease diagnosis or risk prediction by integrating clinical risk factors with examination findings.³⁷ To the best of our knowledge, this is the first study to combine CRF and 2 imaging modalities with distinct measurement principles to diagnose FFKC using ML. Although diagnostic models based on single risk factors exhibit modest

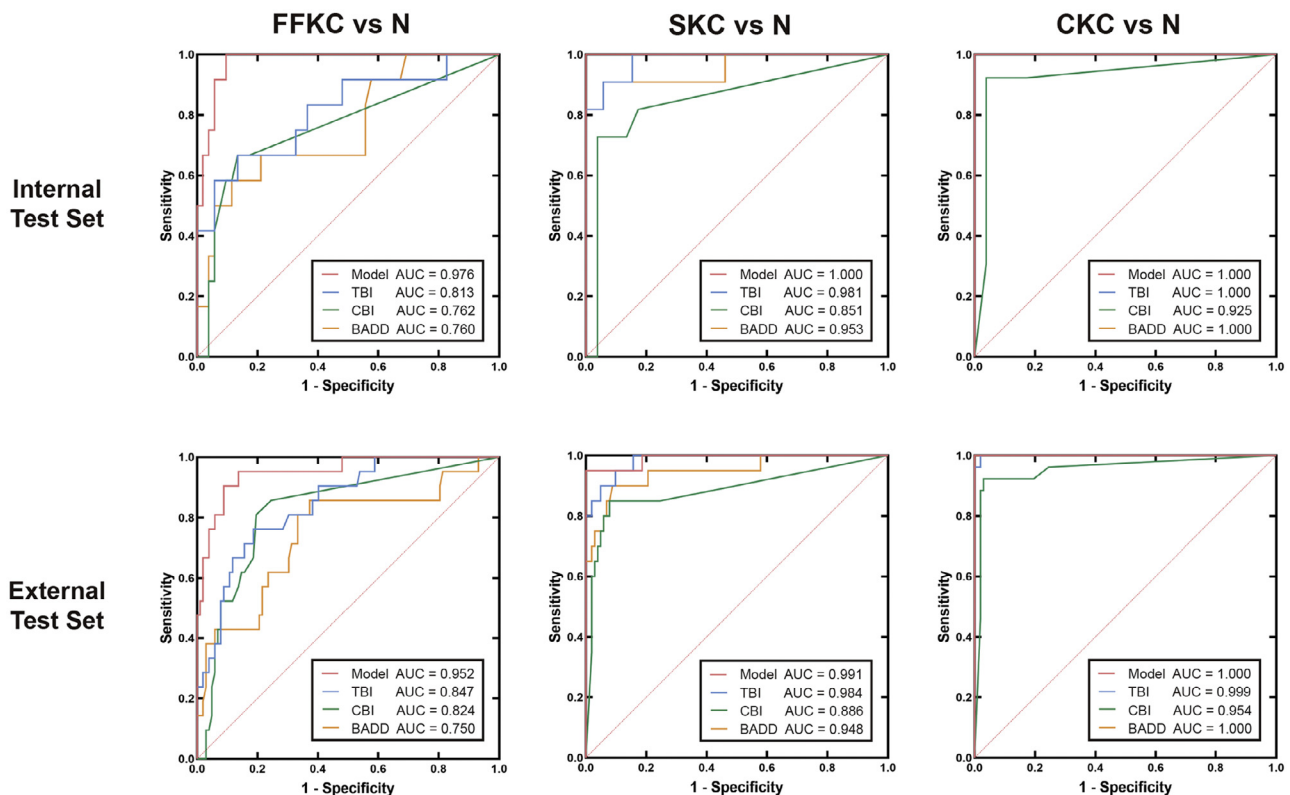


FIGURE 6. Parameter and model diagnostics in internal and external test sets. AUC = area under the curve; BADD = Belin-Ambrósio deviation index; CBI = Corvis biomechanical index; CKC = clinical keratoconus; FFKC = forme fruste keratoconus; N = normal corneas; SKC = subclinical keratoconus; TBI = tomographic and biomechanical index.

performance (CatBoost AUROC: 0.856), integrating CRF into ML frameworks incorporating examination parameters improved the CatBoost model's AUROC from 0.873 to 0.963. This highlights the value of combining CRF with multimodal data to address diagnostic gaps and enable comprehensive risk stratification. Feature selection reduced the original set of 111 variables to 75, further optimizing the performance of the CatBoost model. The refined model achieved AUROC values of 0.975, 0.976, and 0.952, with corresponding Brier scores of 0.055, 0.065, and 0.061 on the validation, internal test, and external test sets, respectively. These results exceeded the diagnostic performance of conventional tomographic and biomechanical indices, including the BADD, CBI, and TBI, while also demonstrating good model calibration. While the model demonstrated near-perfect performance for SKC and CKC, its effectiveness did not significantly differ from existing parameters, likely because traditional indices already provide strong discrimination for these advanced stages.

To address class imbalance resulting from the limited number of FFKC and SKC cases during model development, we employed SMOTE, a widely used technique that generates synthetic minority class samples by linearly interpolating between an existing sample and its k -nearest neighbors.³⁸ Unlike random oversampling, SMOTE lowers

the risk of overfitting by creating novel data points rather than duplicating existing ones, thereby reducing classifier bias toward the majority class and enhancing model sensitivity and performance. However, it does not consider the underlying data distribution or the quality of generated samples, which may introduce noise or redundancy. Additionally, SMOTE can exacerbate intra-class imbalance and perform poorly in datasets with overlapping class boundaries, potentially blurring decision margins.³⁹ Despite these drawbacks, our application of SMOTE yielded consistent performance across validation, internal, and external test sets, indicating minimal overfitting or noise. Nevertheless, we recognize its limitations and emphasize the need for future studies with larger, more balanced datasets, particularly with an increased number of FFKC cases, to improve statistical power and model robustness.

Following this preprocessing step, 6 traditional ML algorithms were evaluated. A central challenge in computational model development is achieving robustness while minimizing unnecessary complexity.^{40,41} Traditional ML models address this by optimizing accuracy and interpretability while mitigating overfitting, enhancing generalizability, and reducing computational demands. Their efficiency in low-resource settings further supports their clinical applicability.

Logistic regression was chosen as the baseline linear model due to its simplicity, robustness, and widespread use in clinical research. However, its reliance on linear assumptions and sensitivity to outliers may limit its performance in complex, nonlinear scenarios.⁴² SVMs, which are widely used for binary classification, were included for their strong performance in high-dimensional spaces and ability to model complex decision boundaries. Nevertheless, SVMs can be computationally intensive, particularly with large or nonlinearly separable datasets.⁴³ Tree-based algorithms, such as RF and gradient boosted decision trees (GBDTs), are well-suited for structured tabular data and often outperform neural networks in such contexts.⁴⁴ Unlike RF, which builds trees independently and aggregates predictions via majority voting, GBDTs construct trees sequentially, with each iteration correcting the errors of the previous one. This enables more effective learning from misclassified instances, particularly those from minority classes. When combined with SMOTE, GBDTs benefits further from their adaptive learning mechanisms, which refine decision boundaries and enhance the utilization of synthetic samples.⁴⁵ Consequently, the XGBoost model with SMOTE outperformed the RF-SMOTE combination in both previous studies⁴⁵ and our analysis. Among GBDTs, XGBoost incorporates regularization techniques to prevent overfitting and handles sparse data efficiently. LightGBM is optimized for large datasets through its leaf-wise growth strategy and memory efficiency, enabling faster training. However, its conversion of categorical variables into numerical values can lead to information loss.⁴⁶ In contrast, CatBoost natively processes categorical variables using advanced encoding methods, requires minimal preprocessing, and employs ordered boosting and symmetric trees to reduce overfitting. Its ability to model complex feature interactions makes it particularly suitable for datasets containing both categorical and numerical features.⁴⁷ Given its superior accuracy and robustness, CatBoost was selected as the optimal model for this study.

Importantly, once the CRF, tomographic, and biomechanical indices were integrated, all evaluated ML models demonstrated strong diagnostic performance, with no statistically significant differences observed. This finding highlights the critical importance of multi-domain feature integration in enhancing diagnostic accuracy, regardless of the specific ML algorithm employed.

Looking forward, the inclusion of genetic and molecular biomarkers alongside inspection and clinical data may further enhance the early detection of KC.⁴⁸ The disease is increasingly recognized as a polygenic disorder, and genome-wide association studies have identified numerous genetic variants linked to KC-related endophenotypes.⁴⁹ These findings have enabled the development of polygenic risk scores (PRS) to quantify individual genetic susceptibility. Although the current predictive performance of PRS remains modest (AUROC: 0.738) and not yet suit-

able for clinical application, this limitation likely reflects the complex interplay between genetic and environmental factors.⁵⁰ Integrating PRS with clinical and environmental risk factors may yield a more comprehensive understanding of disease onset and improve the diagnostic performance of ML models, particularly in KC subtypes where genetic factors play an important role. Among currently emerging technologies, Brillouin microscopy stands out for its ability to detect localized biomechanical alterations and has shown promise in identifying various stages of KC. Incorporating such advanced modalities into ML frameworks may further enhance model performance.⁵¹ Future research should pursue multimodal approaches that integrate genetic, molecular, environmental, and novel diagnostic data to improve the accuracy and generalizability of KC detection.

• **LIMITATIONS:** The study's relatively short duration may have limited the observation of long-term KC progression. Longitudinal studies with extended follow-up are necessary to track disease development and evaluate the model's performance in detecting early KC. While the model was tested using an external dataset, its performance could vary across populations from different geographic regions due to KC prevalence or risk factor variations.³⁵ As such, further validation involving more diverse ethnic groups and broader clinical settings is essential to confirm the model's generalizability.

CONCLUSIONS

Combining CRF, Pentacam, and Corvis ST features proved highly effective in detecting and diagnosing KC, particularly FFKC. This simple and practical multimodal ML-based diagnostic system can play a vital role in identifying patients at risk for corneal ectasia and improving the safety of laser vision correction surgeries.

CREDIT AUTHORSHIP CONTRIBUTION STATEMENT

Kaiyue Du: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Rongmei Peng:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Yueguo Chen:** Resources, Project administration, Methodology, Data curation, Conceptualization. **Bowei Yuan:** Writing – original draft, Visualization, Validation, Investigation, Formal analysis, Data curation. **Haoran Wu:** Visualization, Validation, Software, Resources, Methodology, Formal analysis. **Tiehong Chen:** Resources, Investi-

gation, Data curation. **Jianing Zhu:** Investigation, Data curation. **Xunshan Zu:** Resources, Investigation, Data curation. **Jiaojiao Wang:** Resources, Investigation, Data curation. **Jing Cui:** Resources, Investigation, Data curation.

Liang Han: Writing – review & editing, Supervision. **Jing Hong:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

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REFERENCES

- Mas Tur V, MacGregor C, Jayaswal R, O'Brart D, Maycock N. A review of keratoconus: diagnosis, pathophysiology, and genetics. *Surv Ophthalmol*. 2017;62(6):770–783. doi:10.1016/j.survophthal.2017.06.009.
- Warrak EL, Serhan HA, Ayash JG, et al. Long-term follow up of intracorneal ring segment implantation in 932 keratoconus eyes. *J Fr Ophthalmol*. 2020;43(10):1020–1024. doi:10.1016/j.jfo.2020.03.019.
- Vega-Estrada A, Alió JL, Brenner LF, Burguera N. Outcomes of intrastromal corneal ring segments for treatment of keratoconus: five-year follow-up analysis. *J Cataract Refract Surg*. 2013;39(8):1234–1240. doi:10.1016/j.jcrs.2013.03.019.
- Ferdi AC, Nguyen V, Gore DM, Allan BD, Rozema JJ, Watson SL. Keratoconus natural progression: a systematic review and meta-analysis of 11529 eyes. *Ophthalmology*. 2019;126(7):935–945. doi:10.1016/j.ophtha.2019.02.029.
- Giri P, Azar DT. Risk profiles of ectasia after keratorefractive surgery. *Curr Opin Ophthalmol*. 2017;28(4):337–342. doi:10.1097/ICU.0000000000000383.
- Henriquez MA, Hadid M, Izquierdo L. A systematic review of subclinical keratoconus and forme fruste keratoconus. *J Refract Surg*. 2020;36(4):270–279. doi:10.3928/1081597x-20200212-03.
- Ambrósio R, Lopes BT, Faria-Correia F, et al. Integration of scheimpflug-based corneal tomography and biomechanical assessments for enhancing ectasia detection. *J Refract Surg*. 2017;33(7):434–443. doi:10.3928/1081597x-20170426-02.
- Lopes BT, Ramos IC, Salomão MQ, et al. Enhanced tomographic assessment to detect corneal ectasia based on artificial intelligence. *Am J Ophthalmol*. 2018;195:223–232. doi:10.1016/j.ajo.2018.08.005.
- Vandevenne MM, Favuzza E, Veta M, et al. Artificial intelligence for detecting keratoconus. *Cochrane Database Syst Rev*. 2023;11(11):CD014911. doi:10.1002/14651858.CD014911.pub2.
- Cao K, Verspoor K, Sahebzada S, Baird PN. Accuracy of machine learning assisted detection of keratoconus: a systematic review and meta-analysis. *J Clin Med*. 2022;11(3):478. doi:10.3390/jcm11030478.
- Castro-Luna G, Jiménez-Rodríguez D, Castaño-Fernández AB, Pérez-Rueda A. Diagnosis of subclinical keratoconus based on machine learning techniques. *J Clin Med*. 2021;10(18):4281. doi:10.3390/jcm10184281.
- Gui Y, Wang S, He Y, Zhang S, Zhang Y. Progress of corneal morphological examination combined with biomechanical examination in preoperative screening for keratorefractive surgery. *Indian J Ophthalmol*. 2023;71(6):2369–2378. doi:10.4103/ijo.IJO_1377_22.
- Hashemi H, Heydarian S, Hooshmand E, et al. The prevalence and risk factors for keratoconus: a systematic review and meta-analysis. *Cornea*. 2020;39(2):263–270. doi:10.1097/ico.0000000000002150.
- Yang K, Li D, Xu L, Pang C, Zhao D, Ren S. Independent and interactive effects of eye rubbing and atopy on keratoconus. *Front Immunol*. 2022;13:999435. doi:10.3389/fimmu.2022.999435.
- Eliasi E, Bez M, Megreli J, et al. The association between keratoconus and body mass index: a population-based cross-sectional study among half a million adolescents. *Am J Ophthalmol*. 2021;224:200–206. doi:10.1016/j.ajo.2020.11.021.
- Ahmad TR, Kong AW, Turner ML, et al. Socioeconomic correlates of keratoconus severity and progression. *Cornea*. 2023;42(1):60–65. doi:10.1097/ICO.0000000000002993.
- Jurkiewicz T, Marty A-S. Correlation between keratoconus and pollution. *Ophthalmic Epidemiol*. 2021;28(6):495–501. doi:10.1080/09286586.2021.1879173.
- Mazharian A, Panthier C, Courtin R, et al. Incorrect sleeping position and eye rubbing in patients with unilateral or highly asymmetric keratoconus: a case-control study. *Graefes Arch Clin Exp Ophthalmol*. 2020;258(11):2431–2439. doi:10.1007/s00417-020-04771-z.
- Cheng W-Y, Yang S-Y, Huang X-Y, Zi F-Y, Li H-P, Sheng X-L. Identification of genetic variants in five Chinese families with keratoconus: pathogenicity analysis and characteristics of parental corneal topography. *Front Genet*. 2022;13:978684. doi:10.3389/fgene.2022.978684.
- Awad EA, Abou Samra WA, Torky MA, El-Kannishy AM. Objective and subjective diagnostic parameters in the fellow eye of unilateral keratoconus. *BMC Ophthalmol*. 2017;17(1):186. doi:10.1186/s12886-017-0584-2.
- Rabinowitz YS, Rasheed K. KISA% index: a quantitative videokeratography algorithm embodying minimal topographic criteria for diagnosing keratoconus. *J Cataract Refract Surg*. 1999;25(10):1327–1335. doi:10.1016/s0886-3350(99)00195-9.
- Ferrari G, Rama P. The keratoconus enigma: a review with emphasis on pathogenesis. *Ocul Surf*. 2020;18(3):363–373. doi:10.1016/j.jtos.2020.03.006.
- Sun R, Wang X, Jiang H, et al. Prediction of 30-day mortality in heart failure patients with hypoxic hepatitis: development

- and external validation of an interpretable machine learning model. *Front Cardiovasc Med*. 2022;9:1035675. doi:10.3389/fcvm.2022.1035675.
24. Friedman J, Hastie T, Tibshirani R. Regularization paths for generalized linear models via coordinate descent. *J Stat Softw*. 2010;33(1):1–22. doi:10.18637/jss.v033.i01.
 25. Erdoğan S, Gülhan OT. Alternative confidence interval methods used in the diagnostic accuracy studies. *Comput Math Methods Med*. 2016;2016:7141050. doi:10.1155/2016/7141050.
 26. DeLong ER, DeLong DM, DL Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*. 1988;44(3):837–845. doi:10.2307/2531595.
 27. Steinberg J, Siebert M, Katz T, et al. Tomographic and biomechanical scheimpflug imaging for keratoconus characterization: a validation of current indices. *J Refract Surg*. 2018;34(12):840–847. doi:10.3928/1081597x-20181012-01.
 28. Miao Y-Y, Ma X-M, Qu Z-X, et al. Performance of corvis ST parameters including updated stress-strain index in differentiating between normal, forme-fruste, subclinical, and clinical keratoconic eyes. *Am J Ophthalmol*. 2024;258:196–207. doi:10.1016/j.ajo.2023.10.015.
 29. Vinciguerra R, Ambrósio R, Roberts CJ, Azzolini C, Vinciguerra P. Biomechanical characterization of subclinical keratoconus without topographic or tomographic abnormalities. *J Refract Surg*. 2017;33(6):399–407. doi:10.3928/1081597x-20170213-01.
 30. Luz A, Lopes B, Hallahan KM, et al. Enhanced combined tomography and biomechanics data for distinguishing forme fruste keratoconus. *J Refract Surg*. 2016;32(7):479–494. doi:10.3928/1081597X-20160502-02.
 31. Huo Y, Chen X, Cao H, Li J, Hou J, Wang Y. Biomechanical properties analysis of forme fruste keratoconus and subclinical keratoconus. *Graefes Arch Clin Exp Ophthalmol*. 2023;261(5):1311–1320. doi:10.1007/s00417-022-05916-y.
 32. Hashemi H, Doroodgar F, Niazi S, Khabazkhoob M, Heidari Z. Comparison of different corneal imaging modalities using artificial intelligence for diagnosis of keratoconus: a systematic review and meta-analysis. *Graefes Arch Clin Exp Ophthalmol*. 2024;262(4):1017–1039. doi:10.1007/s00417-023-06154-6.
 33. Ferreira-Mendes J, Lopes BT, Faria-Correia F, Salomão MQ, Rodrigues-Barros S, Ambrósio R. Enhanced ectasia detection using corneal tomography and biomechanics. *Am J Ophthalmol*. 2019;197:7–16. doi:10.1016/j.ajo.2018.08.054.
 34. Guo L-L, Tian L, Cao K, et al. Comparison of the morphological and biomechanical characteristics of keratoconus, forme fruste keratoconus, and normal corneas. *Semin Ophthalmol*. 2021;36(8):671–678. doi:10.1080/08820538.2021.1896752.
 35. Kok YO, Ling Tan GF, Loon SC. Review: keratoconus in Asia. *Cornea*. 2012;31(5):581–593. doi:10.1097/ico.0b013e31820cd61d.
 36. Crawford AZ, Zhang J, Gokul A, McGhee CNJ, Ormonde SE. The enigma of environmental factors in keratoconus. *Asia Pac J Ophthalmol*. 2020;9(6):549–556. doi:10.1097/apo.0000000000000334.
 37. Dupulthys S, Dujardin K, Anné W, et al. Single-lead electrocardiogram artificial intelligence model with risk factors detects atrial fibrillation during sinus rhythm. *Europace*. 2024;26(2):euad354. doi:10.1093/europace/euad354.
 38. Chawla NV, Bowyer KW, Hall LO, Kegelmeyer WP. SMOTE: synthetic minority over-sampling technique. *J Artif Intell Res*. 2002;16:321–357. doi:10.1613/jair.953.
 39. Zhang Y, Deng L, Wei B. Imbalanced data classification based on improved random-SMOTE and feature standard deviation. *Mathematics*. 2024;12(11):1709. doi:10.3390/math12111709.
 40. Rajkomar A, Dean J, Kohane I. Machine learning in medicine. *N Engl J Med*. 2019;380(14):1347–1358. doi:10.1056/nejmra1814259.
 41. Watson DS, Krutzinna J, Bruce IN, et al. Clinical applications of machine learning algorithms: beyond the black box. *BMJ*. 2019;364:l886. doi:10.1136/bmj.l886.
 42. Muhsin ZJ, Qahwaji R, Ghafir I, et al. Advances in machine learning for keratoconus diagnosis. *Int Ophthalmol*. 2025;45(1):128. doi:10.1007/s10792-025-03496-4.
 43. Sheykhmousa M, Mahdianpari M, Ghanbari H, Mohammadianesh F, Ghamisi P, Homayouni S. Support vector machine versus random forest for remote sensing image classification: a meta-analysis and systematic review. *IEEE J Sel Topics Appl Earth Observ Remote Sens*. 2020;13:6308–6325. doi:10.1109/jstars.2020.3026724.
 44. Hollmann N, Müller S, Purucker L, et al. Accurate predictions on small data with a tabular foundation model. *Nature*. 2025;637(8045):319–326. doi:10.1038/s41586-024-08328-6.
 45. Imani M, Beikmohammadi A, Arabnia HR. Comprehensive analysis of random forest and XGBoost performance with SMOTE, ADASYN, and GNUS under varying imbalance levels. *Technologies*. 2025;13(3):88. doi:10.3390/technologies13030088.
 46. Odeh A, Al-Haija QA, Aref A, Taleb AA. Comparative study of CatBoost, XGBoost, and LightGBM for enhanced URL phishing detection: a performance assessment. *J Internet Serv Inf Secur*. 2023;13(4):1–11. doi:10.58346/jisis.2023.i4.001.
 47. Ileri K. Comparative analysis of CatBoost, LightGBM, XGBoost, RF, and DT methods optimised with PSO to estimate the number of k-barriers for intrusion detection in wireless sensor networks. *Int J Mach Learn Cybern*. 2025. doi:10.1007/s13042-025-02654-5.
 48. Bui AD, Truong A, Pasricha N, Indaram M. Keratoconus diagnosis and treatment: recent advances and future directions. *Clin Ophthalmol*. 2023;17:2705–2718. doi:10.2147/opth.s392665.
 49. Hardcastle AJ, Liskova P, Bykhovskaya Y, et al. A multi-ethnic genome-wide association study implicates collagen matrix integrity and cell differentiation pathways in keratoconus. *Commun Biol*. 2021;4(1):266. doi:10.1038/s42003-021-01784-0.
 50. Lee SS-Y, Diaz-Torres S, He W, et al. Polygenic prediction of keratoconus and its measures: cross-sectional and longitudinal analyses in community-based young adults. *Am J Ophthalmol*. 2024;265:248–256. doi:10.1016/j.ajo.2024.05.009.
 51. Cao J, Yu Y, Ma Y, Zhou X, Zhao J. Brillouin microscopy: an emerging tool for biomechanical analysis in ophthalmology. *J Refract Surg*. 2025;41(7):e731–e746. doi:10.3928/1081597x-20250513-02.